

A Two-Dimensional Counterexample to Finite-Dimensional C-Approximate Symmetry

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Status: candidate counterexample, likely valid. The constructed two-dimensional norm has local property (P) at every sphere point, but its Birkhoff–James orthogonality is not globally C-approximately symmetric. This disproves Conjecture 3.11 of the source.

1 Source conjecture

Chmieliński, Khurana, and Sain formulate the following on PDF page 16 of *Local approximate symmetry of Birkhoff–James orthogonality in normed linear spaces*, arXiv:2012.08162 [1]:

Let X be a finite-dimensional Banach space. Then the Birkhoff–James orthogonality is C-approximately symmetric in X if and only if the local property (P) holds for each $x \in \text{Ext } B_X$.

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Conjecture 3.11. Let X be a finite-dimensional Banach space. Then the Birkhoff–James orthogonality is C-approximately symmetric in X if and only if the local property (P) holds for each $x \in \text{Ext } B_X$.

The above conjecture is not true in general. In [5, Example 3.7], an infinite-dimensional smooth space (hence the local property (P) holds for each $x \in S_X$), which is not C-approximately symmetric was constructed.

Figure 1: Conjecture 3.11 as rendered from page 16 of the source paper.

2 Definitions

For $x \in S_X$, let

$$J(x) = \{f \in S_{X^*} : f(x) = 1\}.$$

The support-functional characterizations used in the source are

$$\begin{aligned} x \perp_B y &\iff \text{some } f \in J(x) \text{ satisfies } f(y) = 0, \\ x \perp_B^\varepsilon y &\iff \text{some } f \in J(x) \text{ satisfies } |f(y)| \leq \varepsilon \|y\|. \end{aligned}$$

The relation is globally C -approximately symmetric if one $\varepsilon < 1$ works in

$$x \perp_B y \implies y \perp_B^\varepsilon x$$

for all x, y .

The local property (P) holds at $x \in S_X$ when there is no $y \in S_X$ such that $x \perp_B y$ and every $f \in J(y)$ belongs, up to sign, to $J(x)$. Equivalently, whenever $x \perp_B y$, at least one $f \in J(y)$ lies outside $J(x) \cup (-J(x))$.

3 Proof intuition

Pointwise property (P) only asks for a strict reverse support gap for each orthogonal pair; it does not say that these gaps stay uniformly away from one. The construction makes this distinction visible. Its unit sphere has a flat top facet and a smooth curved arc approaching the left endpoint of that facet. Tangents to the arc are parallel to points moving toward the right endpoint of the facet. This gives exact orthogonal pairs whose reverse support evaluations approach one.

At the limiting corners, extra supporting functionals appear. Those extra supports ensure property (P) at the limit, even though they cannot prevent the pre-limit reverse constants from tending to one. The norm is therefore nonpolyhedral and nonsmooth exactly where the compactness argument behind the conjecture loses upper semicontinuity.

4 The counterexample

Define $u : [-1, 1] \rightarrow \mathbb{R}$ by

$$u(t) = \begin{cases} -1 + 2\sqrt{1+t}, & -1 \leq t \leq 0, \\ 1, & 0 \leq t \leq 1, \end{cases}$$

and set

$$K = \{(s, t) \in \mathbb{R}^2 : -1 \leq s \leq 1, -u(-s) \leq t \leq u(s)\}. \quad (1)$$

Let X be $\mathbb{R}^2$ with unit ball K .

Theorem 1. *The two-dimensional Banach space X defined by (1) has local property (P) at every point of S_X . Nevertheless, Birkhoff–James orthogonality in X is not globally C -approximately symmetric. Consequently, X is a counterexample to Conjecture 3.11.*

Proof. Step 1: K is a unit ball. On $(-1, 0)$,

$$u'(s) = \frac{1}{\sqrt{1+s}}, \quad u''(s) = -\frac{1}{2(1+s)^{3/2}} < 0.$$

At zero the one-sided slope drops from 1 to 0. Hence u is concave. The lower boundary $-u(-s)$ is convex, so K is convex. It is compact, centrally symmetric, and contains the origin in its interior. It is therefore the closed unit ball of a norm on $\mathbb{R}^2$.

The boundary is smooth except at

$$A = (0, 1), \quad B = (1, 1), \quad -A, \quad -B.$$

Every smooth sphere point has local property (P), as observed in the source. It remains to check these four corners. Central symmetry reduces this to A and B .

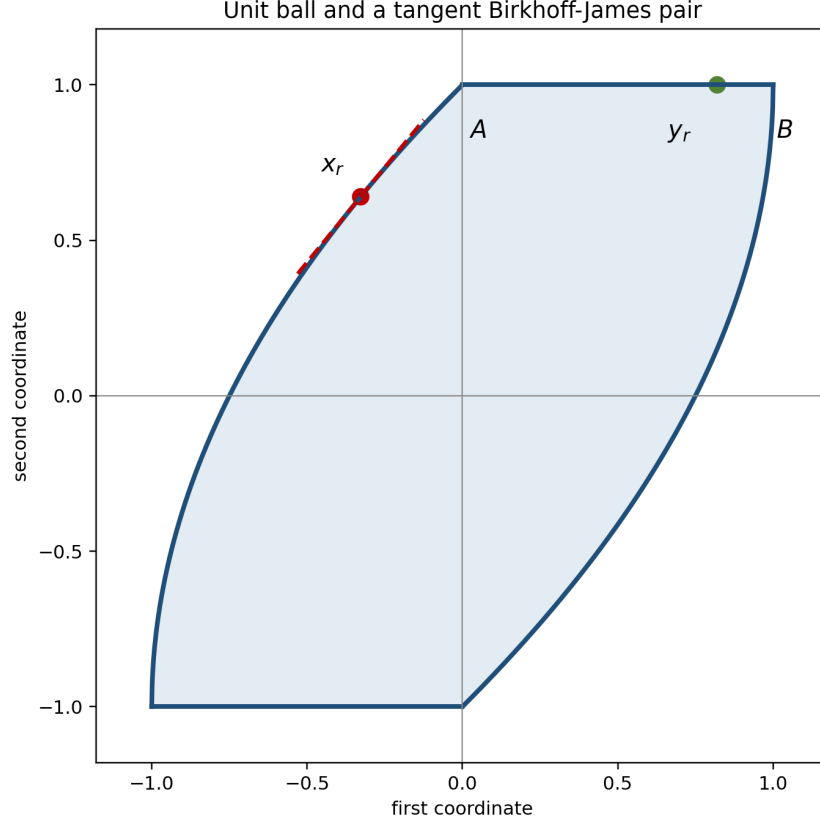


Figure 2: The unit ball. The dashed tangent at x_r is parallel to y_r , while y_r lies in the relative interior of the top facet.

Step 2: property (P) at A . Write the functionals as coordinate pairs. The top facet and the tangent to the square-root arc at A give

$$g = (0, 1), \quad h = (-1, 1), \quad J(A) = \text{conv}\{g, h\} = \{(-s, 1) : 0 \leq s \leq 1\}.$$

Suppose $A \perp_B z$. Up to replacing z by $-z$, some positive multiple of z lies on a ray

$$t = sx, \quad 0 \leq s \leq 1, \text{quad} x > 0.$$

For $s < 1$, this ray meets the relative interior of the smooth right-hand arc

$$\Gamma_R = \{(x, 1 - 2\sqrt{1-x}) : 0 < x < 1\}.$$

Indeed, its defining function has derivative $1/\sqrt{1-x} > 1 \geq s$, begins below the ray, and ends above it. The intersection is unique. The unique supporting functional there cannot lie in $J(A) \cup (-J(A))$: the functionals in $J(A)$ expose either A alone or the top facet $[A, B]$, and their negatives expose the opposite faces.

For $s = 1$, the ray meets B . But $k = (1, 0)$ belongs to $J(B)$ and lies outside $J(A) \cup (-J(A))$. Thus every z with $A \perp_B z$ has a support outside the signed support set of A , proving property (P) at A .

Step 3: property (P) at B . The top facet and the vertical tangent at B give

$$J(B) = \text{conv}\{g, k\} = \{(s, 1 - s) : 0 \leq s \leq 1\}.$$

For $s < 1$, a kernel line of $(s, 1 - s)$ has, in the right half-plane, slope $-s/(1 - s)$. It meets Γ_R at a unique smooth point. As before, the unique support there lies outside $J(B) \cup (-J(B))$, since functionals in $J(B)$ expose only B or the top facet, and the negative functionals expose their opposites. For $s = 1$, the kernel is the vertical axis and meets S_X at $\pm A$; the functional $h \in J(A)$ (or its negative) lies outside $J(B) \cup (-J(B))$. Hence property (P) holds at B , and therefore at all four corners. We have proved property (P) on all of S_X , stronger than the extreme-point hypothesis in the conjecture.

Step 4: no uniform reverse constant. For $1/2 < r < 1$, define

$$x_r = (r^2 - 1, 2r - 1), \quad y_r = (r, 1).$$

Since

$$u(r^2 - 1) = -1 + 2\sqrt{r^2} = 2r - 1,$$

x_r belongs to the smooth square-root arc, while y_r belongs to the relative interior of the top facet. The slope of the arc at x_r is $1/r$, so its tangent vector $(1, 1/r)$ is parallel to $y_r = (r, 1)$. The supporting functional at x_r annihilates y_r , and hence

$$x_r \perp_B y_r.$$

The unique element of $J(y_r)$ is $g = (0, 1)$. By the approximate orthogonality characterization,

$$y_r \perp_B^\varepsilon x_r \iff |g(x_r)| = 2r - 1 \leq \varepsilon.$$

Thus the least reverse constant for this pair is $2r - 1$, which tends to one as $r \uparrow 1$. Given any $\varepsilon < 1$, choosing $r > (1 + \varepsilon)/2$ gives $x_r \perp_B y_r$ but $y_r \not\perp_B^\varepsilon x_r$. No global C-approximate symmetry constant exists. $\square$

5 Verification

The script `code/verify_geometry.py` samples the full boundary, checks the named supporting functionals, verifies five tangent pairs, and stress-tests 2,000 kernel directions at each positive corner. It reports exact named supports, tangent residual 1.11×10^{-16} , and no sampled support entering either forbidden signed corner segment. The largest sampled reverse constant is 0.9998. Full commands and residuals appear in `verification.md`; the computation is not used as proof.

6 Scope and bounded novelty check

The result disproves the finite-dimensional equivalence exactly as stated; it does not affect the source’s theorems for polyhedral or smooth spaces. The counterexample is necessarily outside both classes: it has flat facets, curved arcs, and four nonsmooth junctions.

The run’s cheap indexes were searched by arXiv id, title, and the phrases “C-approximately symmetric”, “local property (P)”, and “Birkhoff–James”. A bounded web/arXiv search used the exact conjecture language and author/title combinations. It found the source and nearby local-symmetry work, but no later resolution or matching construction. This is a bounded check, not an exhaustive novelty claim.

7 Human-review recommendation

This packet is suitable for mathematical review. The highest-value checks are the endpoint cases in Steps 2–3 and the use of the unique top-facet support in Step 4. Until review, retain the status *candidate counterexample, likely valid*.

8 Reference

References

- [1] J. Chmieliński, D. Khurana, and D. Sain, *Local approximate symmetry of Birkhoff–James orthogonality in normed linear spaces*, Results Math. **76** (2021), Article 136; arXiv:2012.08162.